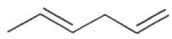
Organic Cume, Sept 9, 2016

1. Rank the following dienes with respect to increasing stability (least stable to most stable) and explain your order.



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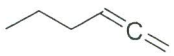

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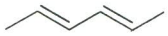

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2. Draw the *s*-trans and *s*-cis conformations of 1,3-butadiene and state which is less stable and why.

3. Addition of hydrogen bromide to 1,3-butadiene gives 3-bromo-1-butene and 1-bromo-2-butene. At room temperature the latter is formed as a major product (56%) whereas at -80 C the former is the major product (81%).

   Show the mechanism that leads to both products. Discuss using a reaction diagram why the product distribution changes with temperature.

4. For 1,3-butadiene, give the HOMO and LUMO molecular orbital probability lobes, along with phases, and rank their energy levels.

5. Consider the reaction of 1,3-butadiene with ethene in a Diels-Alder reaction and show where the overlap of of the molecular orbitals for these reactants occurs.

6. Maleic dialdehyde reacts faster than ethene in a Diels-Alder reaction with 1,3-butadiene. Using a molecular orbital explanation, state why this is the case.

Maleic dialdehyde

7. Give the major product of maleic dialdehyde's reaction with 1,3-cyclopentadiene. Stereochemistry is important. State how you came about this prediction.

8. Give the major Diels-Alder product of the following reaction. Explain how you came about the regiochemistry of your predicted major product.